highly customised FreeBSD system suited perfectly to any unique needs you might have.

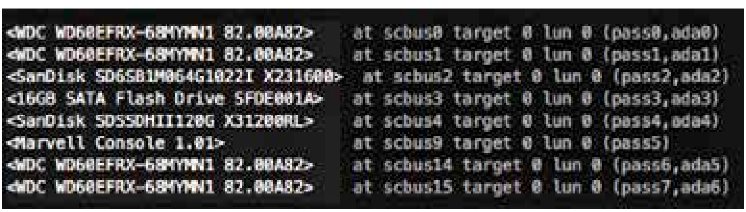
Tuning of disk devices

Disk devices that contain mechanical parts can be a significant factor that brings down the performance of a FreeBSD system. The main resolution to this issue is to use disks that do not contain mechanical parts, for example, solid-state drives. However, when this is not possible, certain files, commands, and variables can be used to tune and optimise disk performance.

The `vfs.vmiodirenable` variable is a binary, with either a 0 (off) or a 1 (on) input. When this is turned off, directories will have only 512 bytes as a part of a buffer cache, regardless of how much system memory is available. On the other hand, when this is turned on, all the system memory is available. If you are working with large files or a significant amount of data, keeping this turned on is recommended.

The `vfs.hirunningspace` variable is used to determine the size of the write I/O that is queued to disks throughout the system. If your system uses multiple disks, it is recommended to keep this set to a maximum of 5 MB.

If your system has several users and multiple idle processes, consider modifying the `vm.swap_idle_enabled` variable. Idle processes generally exert unnecessary pressure on the free memory of a system. You can try to tweak the swapout hysteresis to reduce the memory occupancy of the pages associated with idle processes.



Disks in FreeBSD

Tuning of kernel limits

Every open file or socket in FreeBSD uses a file descriptor. When multiple such files or sockets or fifos are open, thousands of descriptors may be required at any point in time. The `kern.maxfiles` variable in `sysctl` can help you modify the maximum number of file descriptors that can be on a system at any given moment.